

1. Administrative & Study Identification

Full Study Title: A Phase II, Multicenter, Double-Blind, Placebo-Controlled Study to Assess the Efficacy and Safety of R07790121 in Patients With Moderate to Severe Atopic Dermatitis **Short Title/Acronym:** A Study to Assess the Efficacy and Safety of Afimkibart (R07790121) in Participants With Moderate to Severe Atopic Dermatitis **Protocol Number:** CS45570 / 2024-515494-95-00 **NCT Number:** NCT06863961 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** Afimkibart (R07790121 / PF-06480605 / RVT-3101) **Phase of Development:** Phase II **Overall Status:** Active - Recruiting **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy and safety of Afimkibart (R07790121) in participants with moderate to severe atopic dermatitis (AD). **Secondary Objectives:** To assess the long-term safety, tolerability, and sustained clinical effectiveness of Afimkibart over a 48-week period.

2.2 Background & Rationale To address the ongoing medical need for patients with moderate to severe atopic dermatitis who do not adequately respond to current therapies. Afimkibart targets atopic dermatitis in a novel way compared to traditional treatments like topical corticosteroids or calcineurin inhibitors, which mainly focus on reducing inflammation on the skin's surface. Administered subcutaneously, it promises a new systemic mechanism of action (TL1A inhibition) that could lead to improved long-term outcomes with fewer side effects.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Parallel assignment

3.2 Planned Enrollment & Duration Target \$N\$: 160 evaluable patients **Number of Sites:** 69 locations across 8 countries globally (including US, Canada, Europe, and RoW) **Estimated Study Start:** 01-Apr-2025 **Estimated Primary Completion:** 05-Oct-2026 **Estimated Study Completion:** 19-Apr-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years (All genders; Healthy volunteers not accepted). 2. Confirmed diagnosis of moderate to severe Atopic Dermatitis (AD) by a dermatologist according to the Hanifin/Rajka criteria at least 1 year prior to screening. 3. At least once daily use of an additive-free, bland emollient for at least 7 days prior to the baseline visit and during the study.

4.2 Exclusion Criteria 1. Any active infection or other active skin diseases that required treatment with parenteral anti-infectives within 4 weeks or oral anti-infective treatment within 2 weeks prior to baseline. 2. Acquired or congenital immunodeficiency. 3. Systemic therapies that are also used in the treatment of AD, including, but not limited to methotrexate, cyclosporine, azathioprine, mycophenolate mofetil within 4 weeks of the baseline visit and during the study. 4. Topical treatment for AD including, but not limited to topical corticosteroids, topical calcineurin inhibitors, topical PDE-4 inhibitors, prescription moisturizers or moisturizers containing additives such as ceramide, hyaluronic acid, urea or filaggrin within 7 days prior to the baseline visit and during the study.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Afimkibart (R07790121) (Groups I, II, III) or matching placebo. **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** 48 weeks of exposure and follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Additive-free, bland emollient applied at least once daily. **Prohibited:** Systemic AD therapies (e.g., methotrexate, cyclosporine, azathioprine, mycophenolate mofetil). Topical treatment for AD within 7 days prior to baseline and during the study.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Confirmation of Hanifin/Rajka criteria
Baseline	Day 0	Randomization, First Dose (SC Injection), Ensure 7-day prior emollient usage
Interim	Week 16 $\pm$ 3	Primary Efficacy Assessment (EASI-75), Safety Evaluation
End of Study	Week 48	Final Vital Signs, Long-term Efficacy Evaluation, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Percentage of Participants who Achieve Eczema Area and Severity Index-75 (EASI-75) Response ($\geq 75\%$ Improvement from Baseline) at Week 16. **Measurement Method:** Clinical evaluation and scoring by an investigator/dermatologist using the EASI assessment tool.

7.2 Secondary & Exploratory Endpoints

1. Percentage of Participants who Achieve Investigator Global Assessment (IGA) Score of Clear (0) or Almost Clear (1) with ≥ 2 -Grade Improvement at Week 16 and Week 32.
2. Percentage of Participants who Achieve EASI-75 Response at Week 32.
3. Percentage of Participants who Achieve EASI-90 Response.
4. Safety and tolerability endpoints (incidence and severity of adverse events) over 48 weeks.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection via investigator assessment and patient-reported outcomes at scheduled study visits. **Serious Adverse Events (SAEs):** Investigated and reported within 24 hours of clinical site awareness. **Data Monitoring Committee (DMC):** External, independent Data Monitoring Committee to regularly assess safety signals in this Phase II trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy (EASI-75 response); Safety Set for all AE/SAE data. **Sample Size Justification:** Target enrollment of $N = 160$ provides sufficient power ($\alpha = 0.05$, $\text{Power} = 80\%$) to detect a statistically significant difference in EASI-75 response between the Afimkibart group and the placebo group. **Handling of Missing Data:** Non-Responder Imputation (NRI) for primary categorical endpoints (EASI-75); Last Observation Carried Forward (LOCF) or Multiple Imputation for continuous secondary endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Centralized and onsite monitoring visits implemented by Hoffmann-La Roche across all global study sites. **Audit Trail:** Secure Electronic Data Capture (EDC) system with automated audit trails and eCRF locking after query resolution.

11. Study Identifiers & Governance

Primary ID: CS45570 **Registry Number:** NCT06863961 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** A Study to Assess the Efficacy and Safety of Afimkibart (R07790121) in Participants With Moderate to Severe Atopic Dermatitis **Official Regulatory Title:** A Phase II, Multicenter, Double-Blind, Placebo-Controlled Study to Assess the Efficacy and Safety of R07790121 in Patients With Moderate to Severe Atopic Dermatitis

12. Clinical Framework (Atopic Dermatitis Specific)

Primary Condition: Moderate to Severe Atopic Dermatitis **Diagnosis Threshold:** Confirmed via Hanifin/Rajka criteria at least 1 year prior to screening **Medical Methodology:** Biologic therapy combined with continuous baseline emollient use **Optimization Target:** Systemic relief of skin inflammation and itching, targeting an EASI-75 response

13. Study Procedures & Methodology

Management Pathway: Daily additive-free emollient use coupled with scheduled SC biologic administration **Education Protocol (HCP):** Site initiation visits and standardized EASI assessment training **Education Protocol (Patient):** Daily skin care routine compliance and instruction on emollient usage **Quality Audit Mechanism:** Emollient usage diaries, medication accountability logs, and central review of EASI scoring

14. Observation & Follow-Up Schedule

Total Duration: 48 weeks **Onsite Visit Cadence:** Routine intervals extending from baseline through Week 16 (primary endpoint) to Week 48 **Remote Monitoring Frequency:** Continuous patient-reported outcome tracking where applicable **Checklist Review Interval:** Clinical and safety laboratory assessments aligned with dosing visits

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				